

Adopted Levels, Gammas

$Q(\beta^-) = -16891.17$; $S(n) = 17065.34$; $S(p) = 4663.94$; $Q(\alpha) = -6743.9522$ [2021Wa16](#)

$Q(\beta^-)$: Deduced by the evaluator using mass excesses of -1487.17 for ^{34}K measured by [2024Dr01](#); -18378.28980 for ^{34}Ar from [2021Wa16](#). Value from [2021Wa16](#): $Q(\beta^-) = -17160.200$ (syst).

$S(2n) = 32320.618$, $S(2p) = 6940.708$, $Q(\varepsilon) = 6061.796$, $Q(\varepsilon p) = 918.598$ ([2021Wa16](#)).

Identification discovery ([2012Th10](#)): $^{32}\text{S}(^3\text{He}, n)$ at California Institute of Technology ([1966Mi11](#)).

^{34}Ar production:

[2012Zh06](#): ^9Be , $^{181}\text{Ta}(^{40}\text{Ar}, X)$ at $E(^{40}\text{Ar}) = 57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion- ablation, and HIPSE models. Studied target dependence of fragment cross sections.

^{34}Ar radius measurement: [2002Oz03](#).

^{34}Ar mass measurements: [2017Se09](#), [2014Sh14](#), [2002He23](#).

Theoretical calculations: [2024Xa01](#), [2022Zo01](#), [2022Ya19](#), [2021Ma33](#), [2006Or01](#), [2005Ob01](#), [2008Ba08](#) (pygmy E1 resonances), [2005Pa26](#), [2005Sa59](#), [2005Te07](#), [2004Ge02](#), [2001Ot04](#), [1999Ng03](#), [1998La02](#), [1997Ma77](#), [1997Pa38](#), [1995He18](#), [1987Sa19](#), [1977Sh13](#), [1977Sz03](#), [1975Sh10](#).

 ^{34}Ar LevelsCross Reference (XREF) Flags

A	^{35}Ca εp decay (25.7 ms)	E	$^4\text{He}(^{30}\text{S}, \alpha)$: resonances	I	$^{36}\text{Ar}(p, t)$
B	$^1\text{H}(^{34}\text{Ar}, p')$	F	$^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$	J	Coulomb excitation
C	$^1\text{H}(^{35}\text{Cl}, 2n)$	G	$^{32}\text{S}(^3\text{He}, n)$		
D	$^3\text{He}(^{32}\text{S}, n\gamma)$	H	$^{32}\text{S}(^3\text{He}, n\gamma)$		

E(level)	J^π	$T_{1/2}$	XREF	Comments
0.0	0^+	0.84646 s 35	ABCD FGHIJ	$\% \varepsilon + \% \beta^+ = 100$ $T_{1/2}$: from $\beta(t)$ using two component fit to the decay of ^{34}Ar and ^{34}Cl (2020Ia01). Others: 843.8 ms 4 (2006Ia05), 844.5 ms 34 (1974Ha26 , 1972Ha58), 0.74 s 11 (1971Mo27), 0.85 s 10 (1967Ba36), 1.2 s 3 (1966Mi11). J^π : $L=0$ from 0^+ in (p,t) and ($^3\text{He}, n$). Evaluated rms nuclear charge radius $R = 3.3654$ fm 40 (2013An02). XREF: G(2103) J^π : $L=2$ from 0^+ in (p,t) and ($^3\text{He}, n$). $T_{1/2}$: from lifetime $\tau = 2.9\text{E}2$ fs 7: unweighted average of 460 fs 60 from $^3\text{He}(^{32}\text{S}, n\gamma)$ (1985Al18 , DSAM), 330 fs 80 from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Be18 , DSAM), 200 fs 60 from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Gr19 , DSAM), and 150 fs 50 from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1972Ca22 , DSAM). J^π : $L=2$ from 0^+ in (p,t). $T_{1/2}$: from lifetime $\tau = 2.0\text{E}2$ fs 5: weighted average of 280 fs 50 from $^3\text{He}(^{32}\text{S}, n\gamma)$ (1985Al18 , DSAM), 215 fs 70 from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Be18 , DSAM), 130 fs 60 from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Gr19 , DSAM), and 120 fs 65 from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1972Ca22 , DSAM). XREF: G(3900) J^π : $L=0$ from 0^+ in (p,t). Spin=0 from $n\gamma(\theta)$ in $^{32}\text{S}(^3\text{He}, n\gamma)$. $T_{1/2}$: from lifetime $\tau > 600$ fs from $^{32}\text{S}(^3\text{He}, n\gamma)$ (1974Gr19 , DSAM). Other: > 270 fs from $^3\text{He}(^{32}\text{S}, n\gamma)$ (1985Al18 , DSAM).
2091.3 4	2^+	0.20 ps 5	AB D FGHIJ	
3288.5 5	2^+	0.14 ps 3	A D FGHI	
3875.9	0^+	> 0.42 ps	A D FGHI	
4020.6 13	$(2^+)^{\dagger}$		FG I	XREF: G(4050)I(4035)
4129.7 14	$(1, 2)^+{}^{\dagger}$	9×10^1 fs +12-9	D FGH	XREF: G(4150) J^π : spin=2 is favored from $n\gamma(\theta)$ in $^{32}\text{S}(^3\text{He}, n\gamma)$, which did not exclude spin=1, 3, or 4 (1974Gr19); spin=4 ruled out by 840 γ , M1+E2, to 2^+ ; 1^+ assigned in $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar}\gamma)$ based on mirror nucleus ^{34}S .

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Adopted Levels, Gammas (continued)

^{34}Ar Levels (continued)					
E(level)	J^π	$T_{1/2}$	XREF	Comments	
				$T_{1/2}$: from lifetime $\tau=130$ fs +170–130 from $^3\text{He}(^{32}\text{S},n\gamma)$ (1985A118, DSAM). Others: <350 fs from $^{32}\text{S}(^3\text{He},n\gamma)$ (1974Be18, DSAM); >400 fs from $^{32}\text{S}(^3\text{He},n\gamma)$ (1974Gr19, DSAM).	
4515.8 12	$3^{-\dagger}$	0.20 ps 6	B FGHI	J^π : L=3 from 0^+ in $^{32}\text{S}(^3\text{He},n)$; spin=3 from $n\gamma(\theta)$ in $^{32}\text{S}(^3\text{He},n\gamma)$. $T_{1/2}$: from lifetime $\tau=2.9\text{E}2$ fs 8: weighted average of 390 fs 180 from $^{32}\text{S}(^3\text{He},n\gamma)$ (1974Be18, DSAM), 260 fs 80 from $^{32}\text{S}(^3\text{He},n\gamma)$ (1974Gr19, DSAM), and 300 fs 80 from $^{32}\text{S}(^3\text{He},n\gamma)$ (1972Ca22, DSAM).	
4643.8 9	$(4^+)^{\dagger}$		F HI	XREF: H(4631) J^π : 2540 γ , Q, $\Delta J=2$ to 2^+ , 2091 level in ($^{24}\text{Mg}, ^{34}\text{Ar}\gamma$).	
4851.3 8	$(3^+)^{\dagger}$		F		
4865 4			HI	E(level): from $^{32}\text{S}(^3\text{He},n\gamma)$.	
4881.0 12	$(2^+)^{\dagger}$		F i		
4961.8 17	$0^+{}^{\dagger}$		FGHI	XREF: G(4950)H(4967) J^π : L=0 from 0^+ in (p,t) and ($^3\text{He},n$).	
4966.6 11	$2^{(-)^{\dagger}}$		F	J^π : 2875 γ , D, $\Delta J=0$ to 2^+ , 2091 level from $\gamma(\theta)$ in ($^{24}\text{Mg}, ^{34}\text{Ar}\gamma$).	
5060.3 12	$(1^+)^{\dagger}$		F	J^π : from comparison with mirror ^{34}S in ($^{24}\text{Mg}, ^{34}\text{Ar}\gamma$).	
5262 15			HI	XREF: H(5255?) E(level): from $^{36}\text{Ar}(p,t)$. Other: 5255 4 (tentative) from $^{32}\text{S}(^3\text{He},n\gamma)$.	
5316 13	(5^-)		G I	E(level): weighted average of 5310 30 from $^{32}\text{S}(^3\text{He},n)$ and 5317 13 from $^{36}\text{Ar}(p,t)$. J^π : L=(5) from 0^+ in ($^3\text{He},n$).	
5542 4			HI	E(level): from $^{32}\text{S}(^3\text{He},n\gamma)$. Other: 5535 18 from $^{36}\text{Ar}(p,t)$.	
5629 5	2^+		G I	E(level): weighted average of 5620 30 from $^{32}\text{S}(^3\text{He},n)$ and 5629.6 45 from $^{36}\text{Ar}(p,t)$. J^π : L=2 from 0^+ in ($^3\text{He},n$).	
5929 20	0^+		G I	XREF: G(5930) E(level): weighted average of 5930 50 from $^{32}\text{S}(^3\text{He},n)$ and 5929 20 from $^{36}\text{Ar}(p,t)$.	
6050 4	2^+		I	J^π : L=0 from 0^+ in ($^3\text{He},n$).	
6521 15			G I	J^π : L=2 from 0^+ in (p,t). XREF: G(6470) E(level): weighted average of 6470 30 from $^{32}\text{S}(^3\text{He},n)$ and 6525 9 from $^{36}\text{Ar}(p,t)$.	
6641 14			I		
6723 22			I		
6807 4			G I	XREF: G(6820) E(level): from $^{36}\text{Ar}(p,t)$. Other: 6820 40 from $^{32}\text{S}(^3\text{He},n)$.	
6917 5			I		
6.99×10^3 5			G		
7072 8			I		
7276 2			g I	XREF: g(7300)	
7340 11	2^+		g I	XREF: g(7300)I(7340) J^π : L=2 from 0^+ in (p,t).	
7502 16	(2^+)		I	J^π : L=(2) from 0^+ in (p,t).	
7921 18			I		
8166 3			I		
8660 9			I		
8746 8			I		
8899 8			I		
9004 4			I		
9148 22			I		
9226 18			I		

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Adopted Levels, Gammas (continued) ^{34}Ar Levels (continued)

E(level)	J^π	XREF	Comments
9549 16		I	
1.109×10^4 9	(2 ⁺ , 4 ⁺)	E	J^π : from R-Matrix analysis in $^4\text{He}(^{30}\text{S}, \alpha)$: resonances.
1.152×10^4 9	(2 ⁺)	E	J^π : from R-Matrix analysis in $^4\text{He}(^{30}\text{S}, \alpha)$: resonances.
1.208×10^4 10	(2 ⁺)	E	J^π : from R-Matrix analysis in $^4\text{He}(^{30}\text{S}, \alpha)$: resonances.

[†] J^π assignments supported by comparison with mirror nucleus ^{34}S in $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$.

 $\gamma(^{34}\text{Ar})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.	δ	Comments
2091.3	2 ⁺	2091.1 4	100	0.0	0 ⁺	[E2]		B(E2)(W.u.)=11 +4-2 E _γ : weighted average of 2091.3 5 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$, 2090.9 4 from $^{32}\text{S}(^3\text{He}, n\gamma)$, and 2091 2 from Coulomb excitation.
3288.5	2 ⁺	1197.0 5	100 4	2091.3	2 ⁺	M1+E2	+0.12 5	B(M1)(W.u.)=0.079 +22-14; B(E2)(W.u.)=3.1 +34-21 E _γ : weighted average of 1197.5 4 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$ and 1196.5 4 from $^{32}\text{S}(^3\text{He}, n\gamma)$. I _γ : other: 100 6 from $^{32}\text{S}(^3\text{He}, n\gamma)$. Mult., δ: D+Q from $n\gamma(\theta)$ in ($^3\text{He}, n\gamma$); E1+M2 ruled out by RUL.
		3288.9 10	13.9 16	0.0	0 ⁺	[E2]		B(E2)(W.u.)=0.20 +6-4 E _γ : weighted average of 3289.1 10 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$ and 3286 4 from $^{32}\text{S}(^3\text{He}, n\gamma)$. I _γ : weighted average of 14.0 16 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$ and 12 6 from $^{32}\text{S}(^3\text{He}, n\gamma)$.
3875.9	0 ⁺	587.4	<43	3288.5	2 ⁺	[E2]		I _γ : from $^{32}\text{S}(^3\text{He}, n\gamma)$.
		1784.6 8	100	2091.3	2 ⁺	[E2]		E _γ : weighted average of 1784.8 8 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$ and 1782 3 from $^{32}\text{S}(^3\text{He}, n\gamma)$. I _γ : from $^{32}\text{S}(^3\text{He}, n\gamma)$.
4020.6	(2 ⁺)	1930.4 23	65 12	2091.3	2 ⁺			
		4019.8 15	100 12	0.0	0 ⁺			
4129.7	(1,2) ⁺	841.2 13	100 6	3288.5	2 ⁺	M1+E2	+0.52 15	E _γ : unweighted average of 842.5 7 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$ and 839.9 9 from $^{32}\text{S}(^3\text{He}, n\gamma)$. I _γ : from $^{32}\text{S}(^3\text{He}, n\gamma)$. Other: 100 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$. Mult., δ: from $n\gamma(\theta)$ in ($^3\text{He}, n\gamma$); E1+M2, E2+M3, M2+E3 ruled out by RUL.
		2038.3	11 6	2091.3	2 ⁺			I _γ : from $^{32}\text{S}(^3\text{He}, n\gamma)$.
		4129.4	<11	0.0	0 ⁺			I _γ : from $^{32}\text{S}(^3\text{He}, n\gamma)$.
4515.8	3 ⁻	1227.2 12	100 5	3288.5	2 ⁺	[E1]		B(E1)(W.u.)=0.0015 +7-4 E _γ : unweighted average of 1228.4 5 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$ and 1226.0 6 from $^{32}\text{S}(^3\text{He}, n\gamma)$. I _γ : from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$. Other: 100 11

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Adopted Levels, Gammas (continued)

$\gamma(^{34}\text{Ar})$ (continued)							Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult.	
4515.8	3^-	2424.7 22	12.9 32	2091.3	2^+	[E1]	from $^{32}\text{S}(^3\text{He}, n\gamma)$. Mult.: D, $\Delta J=1$ from $\gamma(\theta)$ in $(^{24}\text{Mg}, ^{34}\text{Ar})$; $\Delta\pi=\text{yes}$ from level scheme. B(E1)(W.u.)= $2.6 \times 10^{-5} + I3-8$ I_γ : weighted average of 14.6 32 from $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$ and 11.1 33 from $^{32}\text{S}(^3\text{He}, n\gamma)$.
4643.8	(4^+)	2552.4 8	100	2091.3	2^+	Q	Mult.: Q, $\Delta J=2$ from $\gamma(\theta)$ in $(^{24}\text{Mg}, ^{34}\text{Ar})$.
4851.3	(3^+)	1562.8 7	49.2 34	3288.5	2^+		
		2759.9 12	100 8	2091.3	2^+		
4865		2774	100	2091.3	2^+		
4881.0	(2^+)	1592.5 17	32 5	3288.5	2^+		
		2788.9 19	100 12	2091.3	2^+		
		4881.9 24	82 14	0.0	0^+		
4961.8	0^+	832.1 9	100	4129.7	$(1,2)^+$		
4966.6	$2^{(-)}$	2875.2 10	100	2091.3	2^+	D	Mult.: D, $\Delta J=0$ from $\gamma(\theta)$ in $(^{24}\text{Mg}, ^{34}\text{Ar})$.
5060.3	(1^+)	1771.8 11	100 8	3288.5	2^+		
		5059.9 [‡]	<28	0.0	0^+		
5262		3171	100	2091.3	2^+		
5542		898	54 16	4643.8	(4^+)		I_γ : tentative, from $^{32}\text{S}(^3\text{He}, n\gamma)$.
		1026	100 16	4515.8	3^-		I_γ : tentative, from $^{32}\text{S}(^3\text{He}, n\gamma)$.

[†] From $^{12}\text{C}(^{24}\text{Mg}, ^{34}\text{Ar})$ unless otherwise noted. E_γ values without uncertainties are deduced from level-energy differences.

[‡] Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)